Factors and Multiples

Factors make the number. Multiples come from counting by it.

Every rectangle for 12



1 × 12 2 × 6 3 × 4

Factor pairs of 12: 1 and 12, 2 and 6, 3 and 4.

Factors of 12: 1, 2, 3, 4, 6, 12

Multiples: count by the number

Multiples of 4: 4, 8, 12, 16, 20, …

12 is a multiple of 4, and 4 is a factor of 12.

Prime or composite?



7 has only one rectangle: 1 × 7. It is prime.



8 has more: 2 × 4 and 1 × 8. It is composite.

Your turn!

15 = 1 × ☐ and 3 × ☐

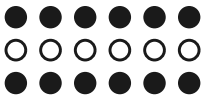
Factors and Multiples: Practice

A factor divides the number with nothing left over.

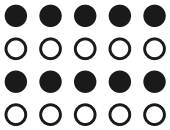
Part 1 Rectangles How many in each row?

1) 

$2 \times \square = 18$

2) 

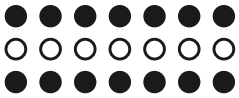
$3 \times \square = 18$

3) 

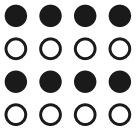
$4 \times \square = 20$

4) 

$2 \times \square = 16$

5) 

$3 \times \square = 21$

6) 

$4 \times \square = 16$

Part 2 Factor pairs Start with 1. Stop when the pairs turn around.

7) $18 = 1 \times \square$ and $2 \times \square$ and $3 \times \square$

Factors and Multiples: Practice (continued)

Part 2 Factor pairs (continued)

8) $20 = 1 \times \square$ and $2 \times \square$ and $4 \times \square$

9) $28 = 1 \times \square$ and $2 \times \square$ and $4 \times \square$

10) $16 = 1 \times \square$ and $2 \times \square$ and $4 \times \square$

11) $30 = 1 \times \square$ and $2 \times \square$ and $3 \times \square$ and $5 \times \square$

12) $21 = 1 \times \square$ and $3 \times \square$

13) $36 = 1 \times \square$ and $2 \times \square$ and $3 \times \square$ and $4 \times \square$ and $6 \times \square$

14) $27 = 1 \times \square$ and $3 \times \square$

Part 3 Is it a factor? Circle yes or no.

15) Is 6 a factor of 42? yes no

16) Is 4 a factor of 30? yes no

17) Is 9 a factor of 48? yes no

18) Is 7 a factor of 56? yes no

Factors and Multiples: Practice (continued)

Part 3 Is it a factor? (continued)

19) **Is 8 a factor of 72?** yes no

20) **Is 3 a factor of 51?** yes no

21) **Is 5 a factor of 32?** yes no

22) **Is 6 a factor of 50?** yes no

23) **Is 4 a factor of 26?** yes no

24) **Is 7 a factor of 84?** yes no

Part 4 All the factors No help this time. List them from smallest to biggest.

25) **Factors of 24:** ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

26) **Factors of 40:** ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

27) **Factors of 32:** ☐ ☐ ☐ ☐ ☐ ☐

28) **Factors of 45:** ☐ ☐ ☐ ☐ ☐ ☐

29) **Factors of 50:** ☐ ☐ ☐ ☐ ☐ ☐

30) **Factors of 60:** ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

Factors and Multiples: Practice (continued)

Part 5 Multiples No help this time. Count by the number.

31) **Multiples of 6: 6 12** ☐ ☐ ☐

32) **Multiples of 9: 9 18** ☐ ☐ ☐

33) **Multiples of 7: 7 14** ☐ ☐ ☐

34) **Multiples of 12: 12 24** ☐ ☐ ☐

35) **Multiples of 8: 8 16** ☐ ☐ ☐

36) **Multiples of 11: 11 22** ☐ ☐ ☐

37) **Multiples of 15: 15 30** ☐ ☐ ☐

38) **Multiples of 25: 25 50** ☐ ☐ ☐

Part 6 Is it a multiple? Circle yes or no.

39) **Is 63 a multiple of 9?** yes no

40) **Is 44 a multiple of 6?** yes no

41) **Is 35 a multiple of 5?** yes no

42) **Is 81 a multiple of 3?** yes no

Factors and Multiples: Practice (continued)

Part 6 Is it a multiple? (continued)

43) **Is 52 a multiple of 8?** yes no

44) **Is 70 a multiple of 4?** yes no

45) **Is 96 a multiple of 8?** yes no

46) **Is 58 a multiple of 7?** yes no

47) **Is 74 a multiple of 9?** yes no

48) **Is 90 a multiple of 6?** yes no

Part 7 Prime or composite? Circle one.

49) **17** prime composite

50) **21** prime composite

51) **9** prime composite

52) **13** prime composite

53) **2** prime composite

54) **27** prime composite

Factors and Multiples: Practice (continued)

Part 7 Prime or composite? (continued)

55) **39** prime composite

56) **11** prime composite

57) **31** prime composite

58) **29** prime composite

59) **51** prime composite

60) **49** prime composite

Part 8 Is it a factor? Circle yes or no.

61) **Is 2 a factor of 41?** yes no

62) **Is 9 a factor of 37?** yes no

63) **Is 9 a factor of 90?** yes no

64) **Is 6 a factor of 66?** yes no

65) **Is 8 a factor of 21?** yes no

66) **Is 8 a factor of 16?** yes no

Factors and Multiples: Practice (continued)

Part 8 Is it a factor? (continued)

67) **Is 6 a factor of 24?** yes no

68) **Is 4 a factor of 69?** yes no

69) **Is 6 a factor of 44?** yes no

70) **Is 8 a factor of 88?** yes no

71) **Is 2 a factor of 86?** yes no

72) **Is 3 a factor of 18?** yes no

73) **Is 9 a factor of 44?** yes no

74) **Is 6 a factor of 18?** yes no

75) **Is 2 a factor of 70?** yes no

76) **Is 7 a factor of 66?** yes no

Factors and Multiples: Practice (continued)

Part 9 Prime or composite?

77) **79** prime composite

78) **42** prime composite

79) **85** prime composite

80) **55** prime composite

81) **26** prime composite

82) **93** prime composite

83) **48** prime composite

84) **74** prime composite

85) **23** prime composite

86) **52** prime composite

87) **83** prime composite

88) **32** prime composite

Factors and Multiples: Practice (continued)

Part 9 Prime or composite? (continued)

89) **24** prime composite

90) **88** prime composite

Part 10 Remember? Problems from earlier packets.

91) **846 ÷ 3 =** ☐

92) **31 × 22 =** ☐

93) **4,568 to the nearest ten =** ☐

94) $\frac{2}{3}$ **of 15 =** ☐

95) **2,705 + 4,198 =** ☐

96) $\frac{5}{6}$ **of 30 =** ☐

Part 11 Word problems Draw the rectangles.

97) A teacher wants 24 desks in equal rows. Will rows of 5 work, or rows of 6?

Number sentence _____

Answer ☐

Draw or show your work

98) Ana packs 36 cookies into bags with the same number in each. Should she put 7 or 9 in a bag?

Number sentence _____

Answer ☐ in a bag

Draw or show your work

99) Buses come every 6 minutes. Does a bus come at 42 minutes after the hour?

Number sentence _____

Answer ☐

Draw or show your work

Factors and Multiples: Practice (continued)

Part 11 Word problems (continued)

100) Leo has 29 tiles. Can he make a rectangle with more than one row?

Number sentence _____

Answer

Draw or show your work

101) Challenge: What is the smallest number that is a multiple of both 4 and 6?

Number sentence _____

Answer

Draw or show your work

Check-In: Factors and Multiples

Show what you know. No help this time. Score: _____ / 9

Part 1 Factor pairs

1) $48 = 1 \times \square$ and $2 \times \square$ and $3 \times \square$ and $4 \times \square$ and $6 \times \square$

2) $40 = 1 \times \square$ and $2 \times \square$ and $4 \times \square$ and $5 \times \square$

Part 2 Is it a factor? Circle yes or no.

3) Is 3 a factor of 60? yes no

4) Is 4 a factor of 90? yes no

Part 3 Is it a multiple? Circle yes or no.

5) Is 48 a multiple of 6? yes no

6) Is 62 a multiple of 6? yes no

Part 4 Prime or composite? Circle one.

7) 53 prime composite

8) 46 prime composite

9) 58 prime composite